

SE1 Module 4 Notes: System Models and Requirements

CS0025 Software Engineering 1. Built from CS0025-M4S0-1, M4S1, and M4S2. The blue boxes are memory aids I made.

How To Use These Notes

Module 4 has two chapters. **Chapter 1** is modeling: system models, architecture, business process models, design diagrams, and **data flow diagrams (DFDs)**. **Chapter 2** is system requirements and the **fact-finding techniques** used in requirements analysis. Expect to be asked to name symbols, name the kinds of models, and match each fact-finding technique to its description.

Chapter 1: System Models

What Is A Model?

A **model** is a description where details are removed in a systematic way and simplified so it is easy to understand. **Modeling a system** means identifying its **characteristics, states or statuses, and behavior** using a notation. **Systems modeling** is the interdisciplinary study of using models or diagrams to conceptualize and construct systems in business and IT development.

The Four Kinds Of Modeling

- Functional Modeling
- System Architecture
- Business Process Modeling
- Enterprise Modeling

Memory Aid

F-A-B-E: Functions Always Benefit Enterprises.

Order goes from small to big: a **function**, the whole **architecture**, the **business** processes, the entire **enterprise**.

Functional Modeling

A structured representation of the **functions, activities, actions, processes, and operations** inside the system. It is a **graphical representation of an enterprise's function within a defined scope**. Purposes: describe functions and processes, help discover information needs, help identify opportunities, and set a basis for working out product and service costs.

The functional perspective is one perspective of business process modeling; the others are **behavioural, organizational, and informational**. It focuses on the **dynamic process**. The main concept is the **process**: a function, transformation, activity, action, or task.

Four Basic Elements Of A Functional Model

Element	Meaning	Shape
Process	Transformation from input to output	Circle or square with a number sequence
Store	A data collection or material	Rectangle, usually with the right side open
Flow	Movement of data or material	Arrow or flow line
External Entity	Outside the modeled system but interacts with it	Rectangle

Memory Aid

P-S-F-E: Please Stop Fighting, Everyone = Process, Store, Flow, External entity.

Functional Block Diagram And FFBD

- **Functional block diagram:** describes the functions and interrelationships of a system. It shows functions as **blocks**, inputs and outputs as **lines**, the relationships between functions, and functional sequences and paths for matter or signals.
- **Functional Flow Block Diagram (FFBD):** a **multi-tier, time-sequenced, step-by-step** flow diagram of the system's functional flow. It defines detailed operational and support sequences and is widely used in software development. Flow steps may combine hardware, software, personnel, facilities, and procedures.

Memory Aid

FFBD = **F**unctional **F**low: the word **Flow** means time-sequenced steps. A plain functional block diagram is just blocks and lines.

System Architecture

The **conceptual model** that defines the **structure, behavior, and more views** of a system. An architecture description is a **formal description and representation** of a system, organized to support reasoning about its structures and behaviors. A system architecture can consist of **components and sub-systems** that work together to implement the overall system.

Business Process Modeling (BPM)

BPM is the activity of representing the processes of an enterprise so the current process can be **analyzed, improved, and automated**. Typical business objectives: increase process **speed** or reduce cycle time; increase **quality**; reduce **costs** (labor, materials, capital). A decision to invest in it is often motivated by the need to document requirements for an IT project.

Business Model

A framework for creating **economic, social, or other forms of value**. It represents the core aspects of a business: purpose, offerings, strategies, infrastructure, organizational structures, trading practices, and operational processes and policies.

Business Process And Its Three Types

A **business process** is a collection of related, structured activities or tasks with a goal to produce a specific service or product for a particular customer.

Type	What It Does	Examples From The Slide
Management Process	Governs system operation	Corporate governance, strategic management
Operational Process	The core business ; creates the primary value stream	Purchasing, manufacturing, marketing, sales
Supporting Process	Supports the core processes	Accounting, recruitment, technical support

Memory Aid

Boss, Builder, Backup. Management = the boss, Operational = the builder of the main value, Supporting = backup.

BPMN

Business Process Model and Notation is a **graphical representation for specifying business processes in a workflow**. The notation is used in a **Business Process Diagram (BPD)**. Goal: support process management for both **technical and business users** with a notation that is intuitive to business people yet can express complex process meaning.

Enterprise Modeling And Data Modeling

Enterprise modeling is the abstract representation, description, and definition of the **structure, processes, information, and resources** of an identifiable business, government body, or other large organization. It means understanding an organization and improving its performance through creating and analyzing enterprise models, including the business processes and the IT used in them.

Data modeling creates a data model using formal data model descriptions and techniques. It is a technique for **defining business requirements for a database**. A data model normally consists of **entity types, attributes, relationships, integrity rules, and definitions**, and is the starting point for interface or database design.

Memory Aid

Data model parts: **E-A-R-I-D** = Entity types, Attributes, Relationships, Integrity rules, Definitions.

Design Diagrams

A design diagram is a **graphic or visual representation of a structure**. It includes **data flow diagrams, structured charts, decision trees**, and other items such as a **cycle of operations (block diagram)**. It shows where data enters a system, where it is processed, the actions taken, and where data leaves as output.

Eight Uses Of A Design Diagram

Use	Meaning
Communications tool	Convenient way to share ideas among engineers, designers, management, and programmers; a concrete visual for complex plans.
Planning tool	Flexible for planning a new system; shows essence and flow lines; analysts see relationships while still planning.
Overview of the system	Shows important elements and relationships without extra details; a bird's-eye view.
Defines roles	Shows the roles of personnel, workstations, and processes; where data, forms, and reports are generated.
Demonstrates relationships	Shows relationships not obvious in words, among data elements and among parts of the system.
Promotes logical procedures	Forces analysts to spell out every path, branch, and flow; logic errors are easier to see visually.
Facilitates troubleshooting	Acts as a blueprint to diagnose breakdowns and communication problems.
Documents the system	Records the key elements clearly and permanently; eases later changes; helps newcomers learn the system.

Memory Aid

One verb each: **Talk, Plan, See, Assign, Connect, Think, Fix, Record.**

Communicate, Plan, Overview, Roles, Relationships, Logic, Troubleshoot, Document.

Data Flow Diagrams (DFDs)

A DFD is a **graphic illustration that shows the flow of data and logic within a system**. The **shape of the symbol** tells the analyst which operation is performed; **arrows** show the **direction** data flows. Labels inside symbols or next to lines describe the flow and transformation. DFD symbols must not be confused with flowchart symbols.

The Four DFD Symbols

Symbol	Shape	What It Is	Notes
Entity	Rectangle	A person or place that provides an input or accepts an output	Source or sink outside the system
Process	Rounded rectangle	An activity or operation; accepts inputs and passes	Numbered sequentially (1.0, 2.0 ...). Shows

	(sometimes a circle)	outputs	where calculations are made or information is changed
Data Flow	Arrow	Movement of data between entities, processes, and stores	Each arrow carries a data object (a form, a document, even verbal speech)
Data Store	Open rectangle	Where data is kept permanently or temporarily	Manual store = simple rectangle with an open end. Automated store = open rectangle with an extra vertical line near the closed end. Must be labeled

Memory Aid

E-P-F-S: Every Person Files Stuff = Entity, Process, Flow, Store.

Shape hooks: Entity = **box** (a thing), Process = **soft box** (it changes things, rounded), Flow = **arrow**, Store = **open shelf** (open side).

Manual vs automated store: **the extra line means automation** (a bar near the closed end).

Watch Out

Functional model says the store is open on the **right**. DFD slide says an **open rectangle**, and a manual store is a rectangle with an **open end**. If asked which store has the extra vertical line: **automated**.

Eight Guidelines For Drawing DFDs

1. **Do not mix levels of detail** on one chart. Draw several DFDs, each at a different level.
2. Pick **one notation**, Gane and Sarson **or** Yourdon and DeMarco, and use it consistently.
3. Use a **template** for uniform symbols in permanent documentation. Freehand is fine for rough or temporary diagrams.
4. **Connect symbols with flow lines**. An arrowhead at one end shows one-way flow; arrowheads at both ends show two-way flow.
5. **Name and label** all symbols and connectors with descriptive names.
6. **Correlate** symbols to other DFDs using reference numbers.
7. **Desk check** every DFD so each symbol is logically connected and describes the flow correctly.
8. **Label the top** of each DFD with the system name, date prepared, preparer's name, and other information.

Memory Aid

Chunk as **3 + 2 + 3**:

How to draw (1 to 3): levels, one notation, template.

How to connect and name (4 and 5): flow lines, labels.

How to check and file (6 to 8): correlate, desk check, label the top.

Reading The Slide Examples

The example on the slides is a **production scheduling system**. One version is a single box, **PRODUCTION SCHEDULING SYSTEM**, surrounded by three external entities (**Supplier, Customer, Production Department**) with flows such as Price Quotation, Specifications, Purchase Order, Price List, Contact, and Report. That is the **big-picture, one-process view**. The other version breaks the system into six numbered processes (1.0 Estimating of Amount, 2.0 Scheduling of Production, 3.0 Preparing of Inventory Report, 4.0 Preparing of Purchase Order, 5.0, 6.0) with data stores such as **Production File** and **Inventory File**. That is the **detailed view**. This is why guideline 1 says not to mix levels.

Chapter 2: System Requirements And Requirements Analysis

System Requirements

System requirements are the specifications a device must have to use certain hardware or software. Example: a computer may need a specific I/O port for a peripheral, or a smartphone may need a specific operating system to run an app. Most software defines **two sets: minimum and recommended**. Requirements tend to **increase over time** as software needs more power.

Example: a video game may run on the minimum CPU and GPU but performs better (better graphics and faster frame rates, **FPS**) on the recommended hardware.

Typical Software Checklist (6)	Typical Hardware Checklist (4)
Operating System (OS)	Operating System (OS)
Minimum CPU or processor speed	Available ports (USB, Ethernet, etc.)
Minimum GPU or video memory	Wireless connectivity (WiFi)
Minimum system memory (RAM)	Minimum GPU (displays and graphics hardware)
Minimum free storage space	
Audio hardware (sound card, speakers)	

Memory Aid

Software list: **O-C-G-R-S-A** = OS, CPU, GPU, RAM, Storage, Audio.

Hardware list is the short one: **O-P-W-G** = OS, Ports, WiFi, GPU.

Flexible Vs Fixed Requirements

- **Not flexible:** the **operating system** and the **disk space** needed to install the software.
- **Can vary a lot** between minimum and recommended: **CPU, GPU, and RAM**.

Memory Aid

OS and Disk are fixed; CPU, GPU, RAM flex. Fixed = "it either fits or it does not".

Requirements Analysis

Requirements analysis covers all tasks in **investigating, scoping, and defining** a new or altered system. The **first activity is the preliminary investigation**, where **data collection** is very important, using **fact-finding techniques**. It is done by **requirements engineers and business analysts**, together with **systems engineers or software developers**, to identify the needs of a client.

Fact-Finding Techniques

Technique	What It Is	Key Points To Remember
Interviews	Analysts collect information about the	Finds misunderstandings, unrealistic expectations, problems, and resistance to

	current system from potential users	the new system. Time consuming.
Record Inspections (Reviews)	Study existing records	Reports, bills, policy manuals, regulations, standard operating procedures
Questionnaires	Collect data from large groups	Two kinds: open-ended and closed-ended (see below)
Observation	Watch the work being done	A skill the analyst must develop: pick the right person, right place, right information; understand how departments and workflows connect
Research	An advanced fact-finding technique	Get information from published materials or documents to check and verify facts

Memory Aid

I Read Questions, Observe, Research = Interview, Records, Questionnaire, Observation, Research.

Interview = slow but rich. Questionnaire = many people. Record review = paper trail.

Observation = watch. Research = check facts in published sources.

Open-Ended Vs Closed-Ended Questionnaires

	Open-Ended	Closed-Ended
Answers	In the person's own words	Choose from a set of prescribed answers
Used to learn	Feelings, opinions, general experiences, process detail or problems	Specific responses
Type of data	Qualitative	Quantitative
Cost note		Costly because the questions must be printed

Memory Aid

Open = Opinions = Qualitative. Closed = Counts = Quantitative.

One-Page Cram Sheet

Topic	Remember
Four modeling kinds	Functional, System Architecture, Business Process, Enterprise (FABE)
Functional model elements	Process, Store, Flow, External Entity (PSFE)
FFBD	Multi-tier, time-sequenced, step-by-step
Architecture	Conceptual model of structure, behavior, views
Business process types	Management, Operational (core), Supporting
BPMN	Notation for workflows, drawn as a Business Process Diagram
Data model parts	Entity types, attributes, relationships, integrity rules, definitions
Design diagrams include	DFDs, structured charts, decision trees, block diagrams
DFD symbols	Entity (rectangle), Process (rounded, numbered), Flow (arrow), Store (open rectangle)
Automated data store	Open rectangle with an extra vertical line near the closed end
DFD notations	Gane and Sarson, or Yourdon and DeMarco: pick one
Requirements sets	Minimum and recommended
Fixed requirements	OS and disk space
Fact-finding	Interviews, records, questionnaires, observation, research